

Search Results (2022–2025)

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"Automated Compliance Checking in the AEC Industry: A Review of Current State, Opportunities and Challenges","Muhammad Yamusa; Jamiu Dauda; Saheed Ajayi; Abdullahi B. Saka; Adekunle S. Oyegoke; Waris Adebisi; Zainab T. Jagun",2024,"https://doi.org/10.2139/ssrn.5053757","Automated Compliance Checking (ACC) is continuously gaining traction in improving the efficiency and precision of regulatory compliances within the AEC sector. Thus, this research presents a comprehensive review of the current state of ACC, emphasising its application domains, techniques, challenges and opportunities. The review reveals that ACC is currently being applied in multiple domains including building design analysis, energy efficiency, construction safety and fire safety. ACC systems employ techniques such as artificial intelligence (deep learning, machine learning, and natural language processing), graph-based methods, semantic enrichment and representation, and general rule representation analysis. The review identifies technological constraints and integration difficulties as the main challenges facing ACC implementation. The potential opportunities for ACC include integration with enhanced technologies, expanding application domains, collaborations and standardisations. This study addresses existing knowledge gaps and enhances the understanding of ACC's role and impact, steering future research towards innovative approaches and improved implementation strategies ¹ ²","Preprint (SSRN) ³ "

"Semantic web-based automated compliance checking with integration of Finite Element analysis","Panagiotis Patlakas; Ioannis Christovasilis; Lorenzo Riparbelli; Franco K. T. Cheung; Edlira Vakaj",2024,"https://doi.org/10.1016/j.aei.2024.102448","Automatic Compliance Checking (ACC) is a promising response to the challenges involved in meeting building and planning regulations, and is increasingly utilised in the context of Building Information Modelling (BIM) and Industry Foundation Classes (IFC). However, engineers often use computational methods such as Finite Element Analysis (FEA) to solve challenging engineering problems, which have received little attention in the context of ACC and BIM. This work proposes a holistic approach integrating FEA into the IFC schema and utilising a Semantic Web-based system for ACC. A top-level ontology, the Building Compliance Ontology (BCO), is proposed to cover all aspects of compliance in the built environment - one novelty is that it models statements in the building regulations (as opposed to the common approach of modelling building components). To evaluate this process, a real-world case is used: checking compliance of structural members of a modular steel structure with structural design codes. An open-source solver is utilised for FEA, and the IFC model is enriched with its results via an open-source middleware. BCO is extended to structural engineering, and appropriate ACC classes and SHACL checks are implemented. Compliance is checked automatically by running SHACL scripts in a Semantic Web environment ⁴ ⁵","Advanced Engineering Informatics ⁶ "

"Lessons Learned on Adopting Automated Compliance Checking in AEC Industry: A Global Study","Yang Zou; Brian H.W. Guo; Eleni Papadonikolaki; Johannes

Dimyadi; Lei Hou",2023,"<https://doi.org/10.1061/JMENE/MEENG-5051>","Over the last decades, numerous Automated Compliance Checking (ACC) systems have been developed. However, ACC is still not broadly used in practice, and little is known about how it can be better accepted by end users. This paper reports on a multiple-case study to learn valuable lessons from recent attempts to adopt ACC systems worldwide. Eighteen semi-structured interviews were conducted with twenty experts from eight countries, and supplementary data (documents, product information, literature) for each case were collected. The interview data were coded to develop prominent themes, and through cross-case analysis, twelve key variables influencing ACC adoption were identified. Three path models explaining the interrelationships between these variables and ten propositions guiding future ACC adoption were deduced. The results indicate that governments should play an important role in facilitating ACC adoption through funding, policies, and incentives. The study also provides valuable information to software vendors for delivering ACC systems that meet industry needs, and offers guidance for innovation managers to develop appropriate ACC technology adoption plans ⁷ ⁸","Journal of Management in Engineering ⁹ "

"Automated Compliance Checking Situation in Health and Safety Management in UK's Infrastructure Sector","Zijing Zhang; Algan Tezel; Maxwell F. Antwi-Afari; Tala Kasim; Doug Potter",2024,"<https://doi.org/10.35490/EC3.2024.285>","Despite intensive research efforts in developing automated compliance checking (ACC) systems for improved health and safety (H&S) outcomes, existing research has mainly focused on technical aspects and the building sector, while social and organisational aspects of ACC in the infrastructure sector are lacking. To address this gap, this study uses a case study to explore the issues and readiness for implementing ACC in H&S within a design organisation in the UK's infrastructure sector. The findings show low readiness with issues in four main aspects, and suggestions are proposed for improvement. This research contributes by identifying gaps and proposing ways to achieve better readiness for implementing ACC in design organisations in the UK's infrastructure sector ¹⁰ ¹¹","2024 European Conference on Computing in Construction (Proceedings) ¹² "

"Automated Compliance Checking System for Structural Design Codes in a BIM Environment","Wonhui Goh; Jaeguk Jang; Sang I. Park; Bong-Hyuck Choi; Heuiseok Lim; Goangseup Zi",2024,"<https://doi.org/10.1007/s12205-024-1121-5>","This study proposes a framework for developing an automated compliance checking (ACC) system that supports verification of structural design calculations in an openBIM format. The framework consists of four main components: (1) classification of design codes by structural component and limit state; (2) preparation of a structural model according to Industry Foundation Classes (IFC); (3) interpretation of code conditions into a machine-readable rule language; and (4) design of an ACC system with focus on rule flow in the structural design code. The framework is demonstrated with a detailed example ACC system using ifcOWL, SWRL and the Drools inference engine. The ACC system checks the strength and serviceability of a prestressed concrete girder bridge and its components, and it successfully verifies an actual design example. The proposed framework can assist in developing ACC systems in structural engineering and can be customized to meet user requirements and accommodate various data sources and design codes ¹³ ¹⁴","KSCE Journal of Civil Engineering ¹⁵ "

"Rule capture of automated compliance checking of building requirements: a

review", "Zijing Zhang; Ling Ma; Tim Broyd", 2023, "https://doi.org/10.1680/jsmic.23.00005", "In the AEC industry, building designs must be checked against regulations before construction. The traditional manual compliance-checking process is error-prone and time-consuming. Automated compliance checking (ACC) has been proposed as a solution, but the *rule capture* stage remains a crucial bottleneck. Despite many studies, no work has yet synthesized themes and future research opportunities in this area. This review addresses that gap by systematically examining literature and identifying challenges. Findings reveal that interest in automated rule capture has grown in recent years, with more semi-automated methods proposed. However, current rule representation processes lack a sound methodology. Existing representations cannot capture 'unknowns' or 'side effects', struggle with ambiguous rules, and are often limited by specific rule engines or data models. The understanding of rules, their representations, and interrelationships is still insufficient. Further research is needed to address these issues ¹⁶ ¹⁷ ", "Proceedings of the ICE - Smart Infrastructure and Construction ¹⁸ "

"Large Language Model-Driven Code Compliance Checking in Building Information Modeling", "Soumya Madireddy; Lu Gao; Zia Ud Din; Kinam Kim; Ahmed Senouci; Zhe Han; Yunpeng Zhang", 2025, "https://doi.org/10.3390/electronics14112146", "This research addresses the time-consuming and error-prone nature of manual code compliance checking in Building Information Modeling (BIM) by introducing a Large Language Model (LLM)-driven approach to semi-automate this process. The developed system integrates LLMs (ChatGPT, Claude, Gemini, Llama) with Autodesk Revit to interpret building codes, generate Python scripts, and perform semi-automated compliance checks within the BIM environment. Case studies on a single-family house and an office building demonstrated the system's ability to reduce the time and effort required for compliance checks while improving accuracy. It automatically identifies violations (e.g. non-compliant room dimensions, material usage, object placements) by assessing model relationships and generating actionable reports. Compared to manual methods, the system eliminates repetitive tasks, simplifies complex regulations, and ensures reliable adherence to standards. As a comprehensive, adaptable, and cost-effective solution, this LLM-driven approach offers a promising advancement in BIM-based compliance checking with potential applications across diverse regulatory documents ¹⁹ ²⁰ ", "Electronics (MDPI) ²¹ "

"Moving automated compliance checking to the operational phase of the building life-cycle: analysis and feasibility study in the UK", "Thomas Beach; Jonathan Yeung; Ali Ghorghi; Yacine Rezgui", 2024, "https://doi.org/10.1080/15623599.2024.2366727", "Construction regulations ensure a base level of safety for built assets. Formal compliance checks of built assets against regulations occur at defined points in the asset life-cycle, often with limited automation. The push towards digitising and automating site inspections paves the way for a whole life-cycle approach to compliance checking. This paper explores extending current compliance-checking processes (common in design and construction) into the operational phase of buildings. It proposes that the best approach is through automated/semi-automated data capture and analysis. To test feasibility, the paper reviews literature, analyses regulations, and presents a proof-of-concept. Findings show that whole life-cycle compliance checking is increasingly important for monitoring

high-risk elements of the built environment. However, when applied to the operational phase, relying solely on manual inspections is impractical and costly. There are numerous viable use cases for this approach, and technologies exist to support them ²² ²³ ", "International Journal of Construction Management ²⁴ "

"Using Large Language Models for the Interpretation of Building Regulations", "Stefan Fuchs; Michael Witbrock; Johannes Dimyadi; Robert Amor", 2023, "https://arxiv.org/abs/2407.21060", "Compliance checking is an essential part of construction projects. The rapid adoption of Building Information Models (BIM) has created more opportunities for automated compliance checking (ACC). BIM enables sharing of digital building design data for checking against legal requirements, which are typically written in natural language and not machine-readable. Converting regulatory text into a computable form for ACC is complex, costly, and time-consuming. Large Language Models (LLMs) such as GPT-3.5 and GPT-4 can generate coherent text and source code from prompts, suggesting a new way to automate the translation of building regulations into a semantic, computable format. This paper evaluates how well LLMs can translate building regulations into LegalRuleML using few-shot learning. By providing GPT-3.5 a few example translations, the model learns the basic format. Using a system prompt, we further refine the LegalRuleML output and probe the model's built-in domain knowledge. We also explore strategies like chain-of-thought reasoning and self-consistency for this task. As LLMs advance, their improved common sense, logical coherence, and adaptability could significantly support ACC, leading to more efficient and effective compliance checking processes ²⁵ ²⁶ ", "13th Int. Conf. on Engineering, Project and Production Management (EPPM 2023) ²⁷ "

"How Can ChatGPT Help in Automated Building Code Compliance Checking?", "Jiansong Zhang", 2023, "https://doi.org/10.22260/ISARC2023/0011", "One major challenge in fully automating building code compliance checking is extracting and transforming code requirements into computable representations. A semantic rule-based approach has generally outperformed machine learning on this task. With recent advances in AI - notably OpenAI's ChatGPT - there is potential to shift this landscape given ChatGPT's strong ability to understand and generate natural language and code. This paper preliminarily explores using ChatGPT to convert building code requirements into computer-executable code, and compares the results with a cutting-edge semantic rule-based approach. It was found that, compared to the rule-based method, ChatGPT's conversion still has limitations. However, ChatGPT shows great potential to accelerate the implementation and scaling of automated building code compliance checking systems ²⁸ ²⁹ ", "Proc. of the 40th International Symposium on Automation and Robotics in Construction (ISARC 2023) ³⁰ "

"Automated code compliance check of residential buildings for parking regulation in Turkey", "Bayram Ali Temel; Hasan Basri Başağa", 2022, "https://doi.org/10.1680/jmuen.21.00024", "Compliance inspection of a building - following legislated rules during the design phase - is a manual, time-consuming, error-prone and labour-intensive process that relies primarily on 2D drawings. Building Information Modelling (BIM) innovations enable automatic compliance inspection of buildings against relevant regulations. A BIM model (in Industry Foundation Classes format) contains all pertinent 3D geometric and semantic data of a building. Using these BIM/IFC data, this

study developed a program (intended for use by Turkish municipalities, but applicable elsewhere) that automatically checks a building's compliance with area parking requirements. This paper presents the program's development and test results. The program simplifies compliance checking by determining the required parking area size and number of parking spaces according to Turkish residential parking regulations, and by performing the checks accordingly. Although not yet commercially implemented, the program can provide convenience and speed to stakeholders (architects, engineers) during design and review, while also minimizing error rates ³¹ ³² ", "Proceedings of the ICE - Municipal Engineer ³³ "

¹ ² ³ Automated Compliance Checking In The Aec Industry: A Review Of Current State, Opportunities And Challenges by Muhammad Yamusa, Jamiu Dauda, Saheed Ajayi, Abdullahi B. Saka, Adekunle S. Oyegoke, Waris Adebisi, Zainab T. Jagun :: SSRN

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⁴ ⁵ ⁶ Semantic web-based automated compliance checking with integration of Finite Element analysis

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¹⁰ ¹¹ ¹² Automated compliance checking situation in health and safety management in UK's infrastructure sector

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²⁵ ²⁶ ²⁷ EPPM_2023_Arxiv

<https://arxiv.org/pdf/2407.21060>

²⁸ ²⁹ ³⁰ [iaarc.org](http://www.iaarc.org)

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